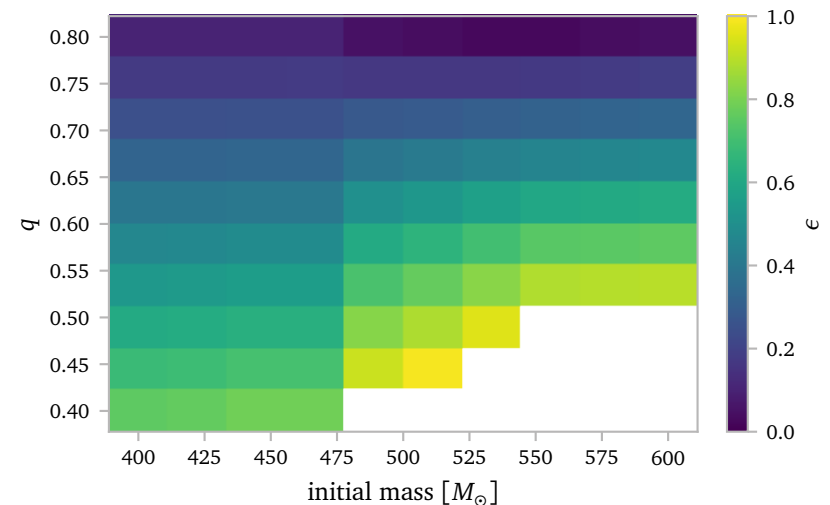


Notes Week 23

a. Grid

Here I will note down my most important findings for the grid that is based on the accretion efficiency required to produce a $1.29 M_{\odot}$ barium star.



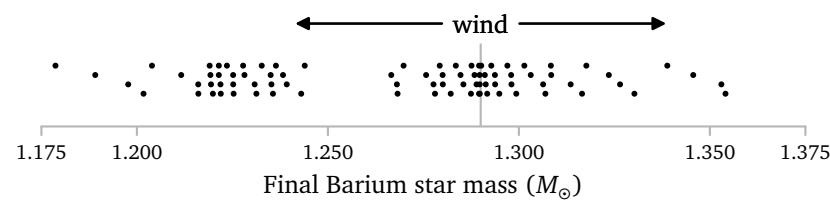
This plot show the ϵ values for the different $R_{\text{Roche lobe}}$ and q combinations. Note that the bottom right 3 squares are empty because they cannot produce a $1.29 M_{\odot}$ barium star, even with 100% accretion. The grid only goes up to $q = 0.8$ because $q = 0.9$ would produce barium stars heavier than $1.29 M_{\odot}$ even without any accretion.

Below I investigate the 3d¹ grid that I simulated last week.

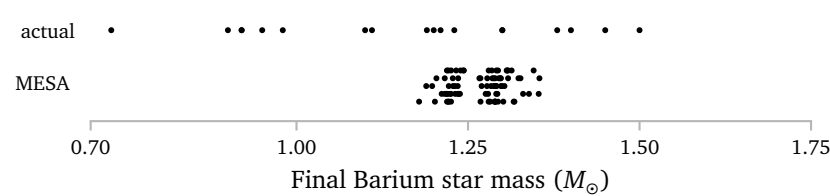
1 5 × 5 × 5

The first thing I want to check is *how* close the final accretor masses are to the $1.29 M_{\odot}$ value.

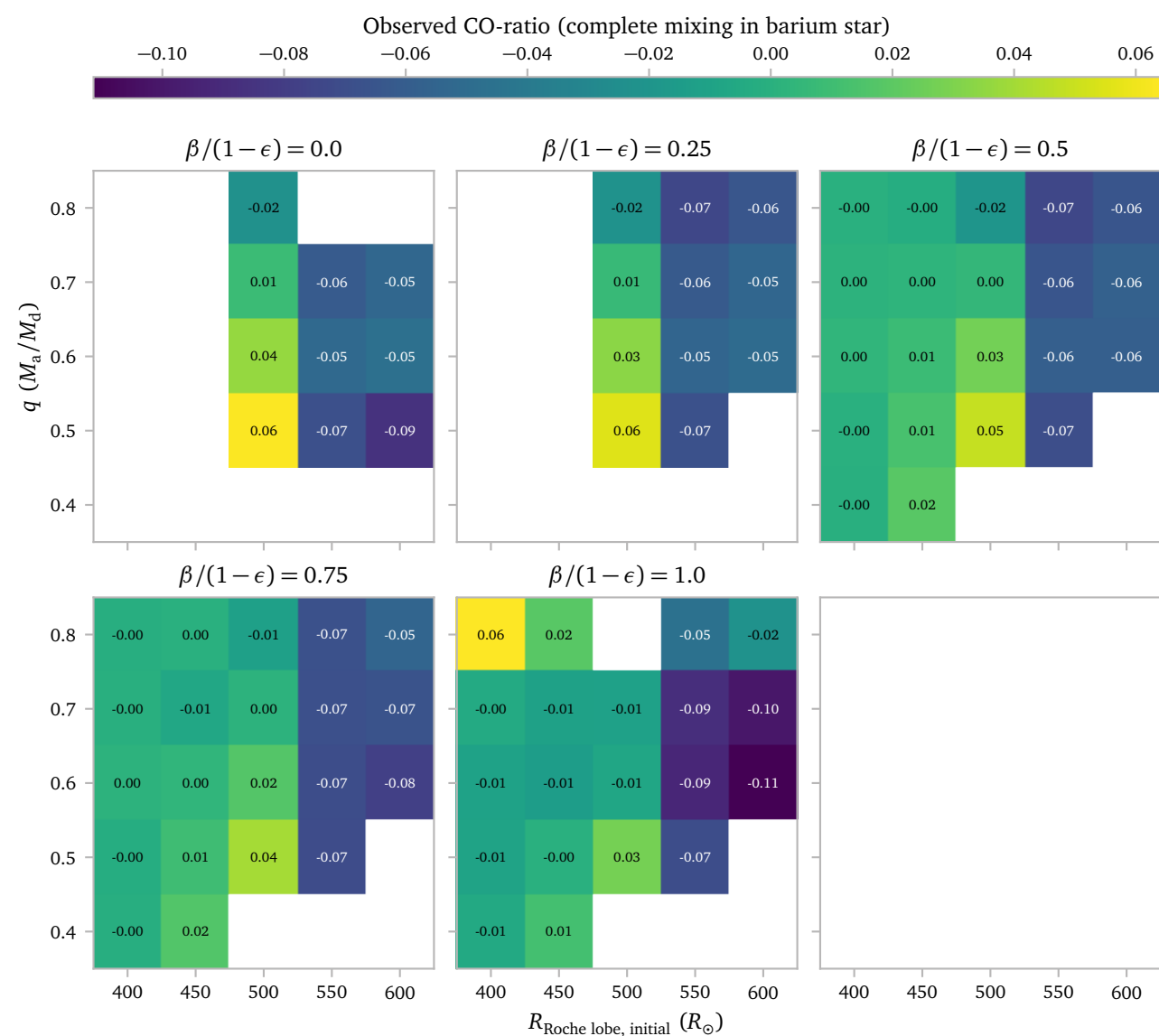
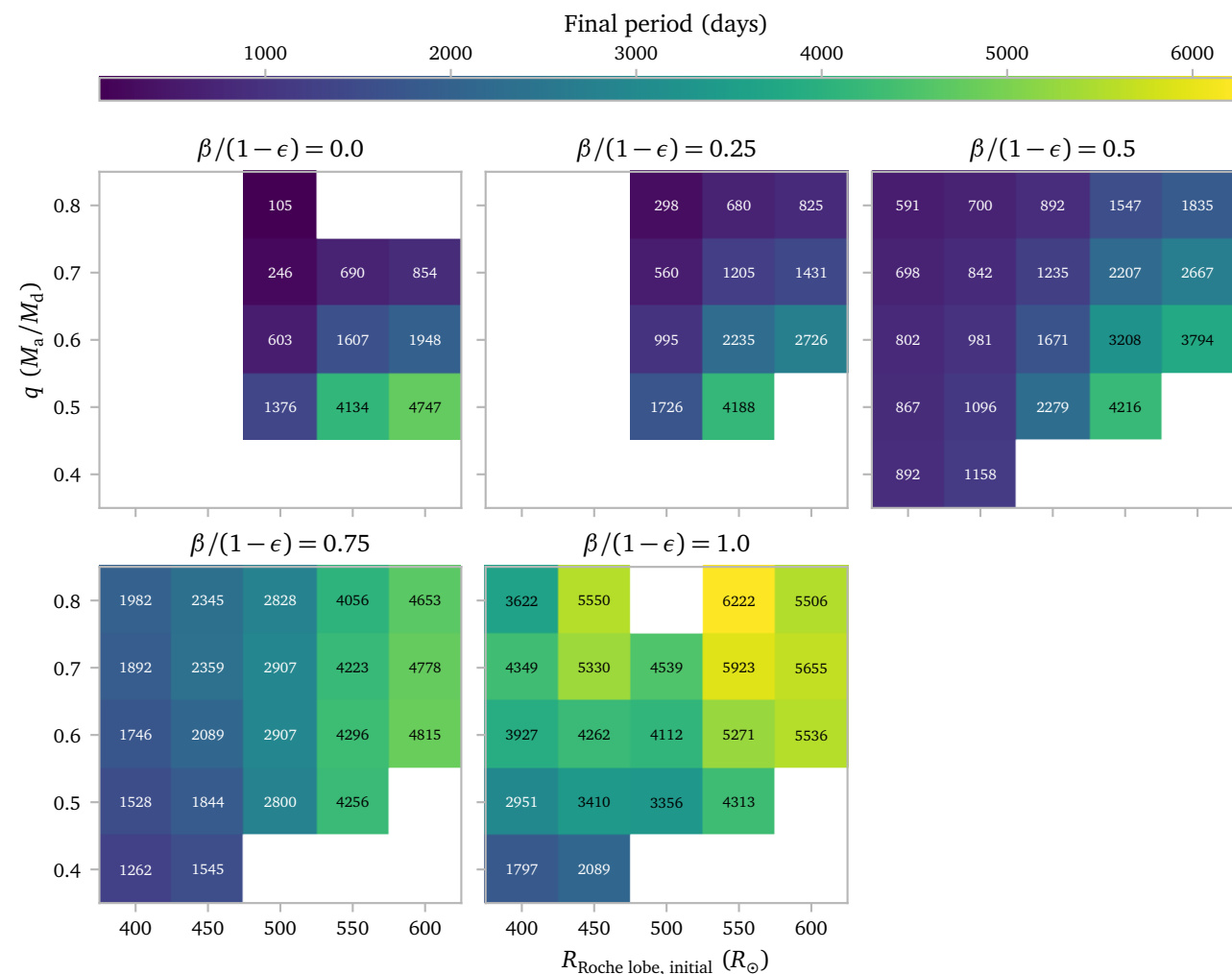
If there would be no wind accretion after the initialization of the models, and they all interact at the predicted time, all models should have a mass of $1.29 M_{\odot}$.



We produce many stars close to the $1.29 M_{\odot}$ final barium mass, but there is a spread with a maximum of about $0.1 M_{\odot}$. This looks quite bad here. Let's compare it with the actual barium star distribution.

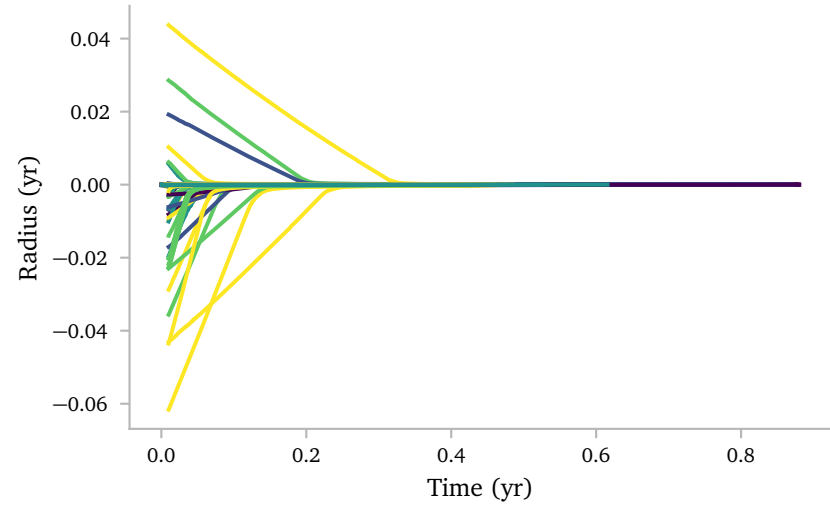


Now we can see that the spread in the final masses of the *MESA* models is quite a bit smaller than the spread of the Barium stars.



b. fix

Some of the models in the grid interact directly after starting.



This is not *necessarily* wrong but it is *better* to start the simulation a bit earlier, so we know for sure the stars would not have interacted at and earlier pulse.

With the refactor of the old grid-generation code, I accidentally forgot to take the model BEFORE the first model that has a radius smaller than $0.9 \times R_{\text{RL}}$. Below, I show the *new* and *corrected* version that does have this feature again.

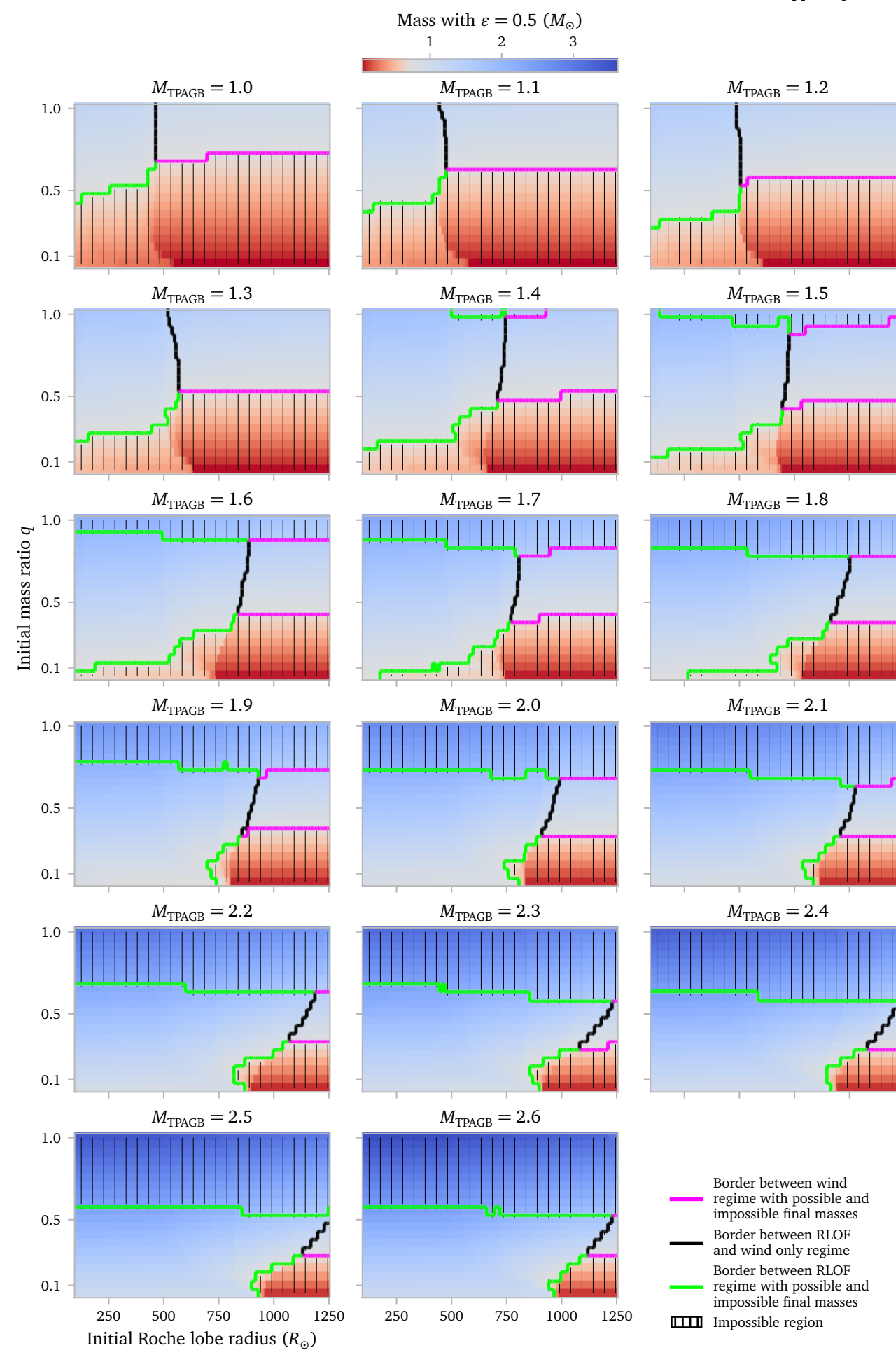
```
def select_model(contact_age, models_dict):  
    candidate = None  
    models = list(models_dict.values())  
  
    for i in range(len(models) - 1, -1, -1):  
        if models[i]["age"] < contact_age:  
            # Return the previous model if it exists  
            if i > 0:  
                return models[i - 1]  
  
    print("no suitable stellar model before contact ->  
    skipping")  
    return candidate
```

c. Suitable MESA models

The masses of the barium dwarfs range from 0.73 to $1.5 M_{\odot}$. If we want to produce barium stars with these masses using MESA, and limit ourselves to a range of mass transfer efficiencies, we can exclude *many* binary configurations.

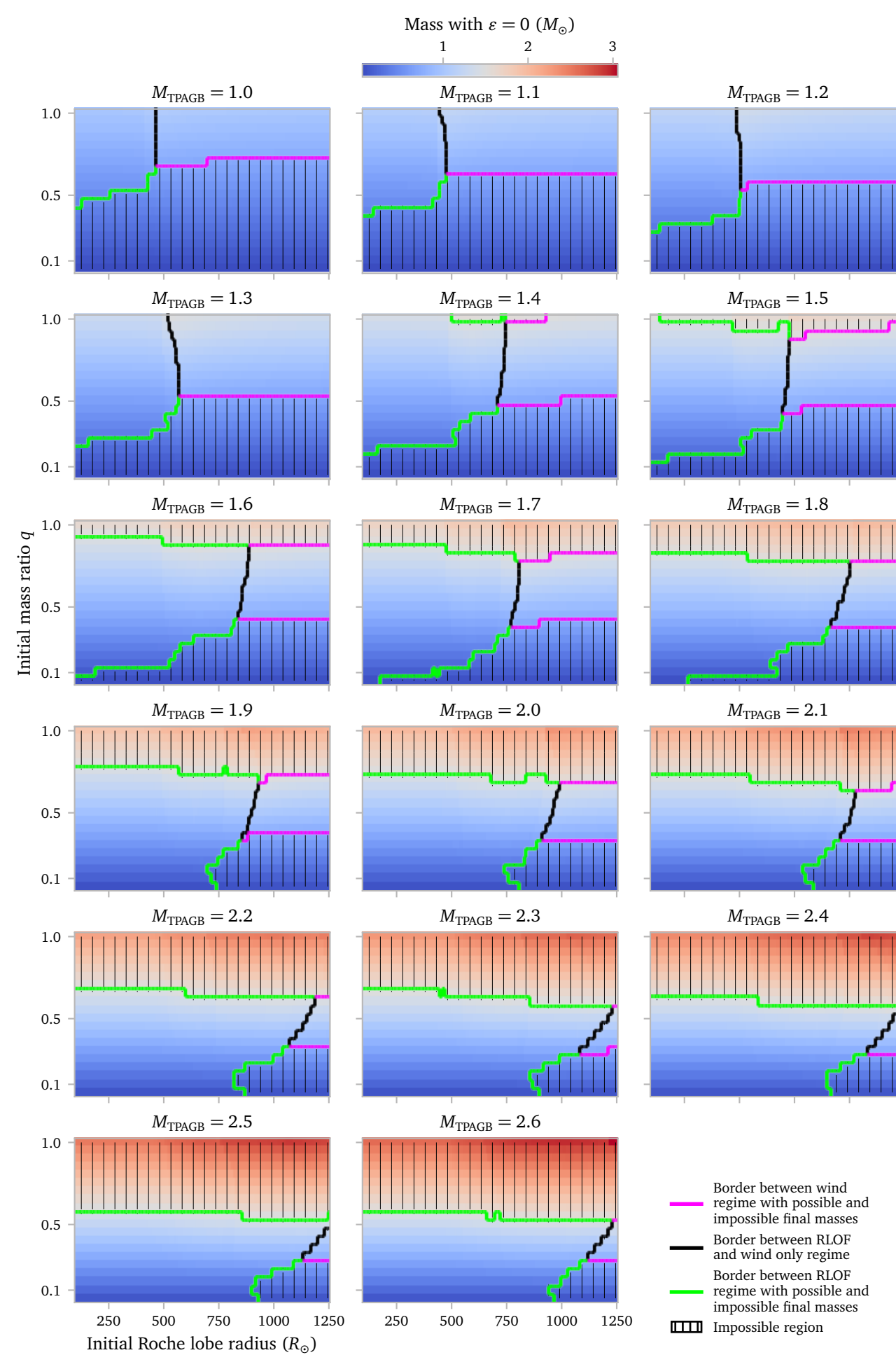
To see this in action, below I plot the mass of the final barium star after RLOF with an accretion efficiency of 0.5 , which in this case, I assume is the *upper* boundary of the efficiency².

² It seems unlikely that for extreme RLOF like what is happening here,



This means we can exclude all of the regions that are *red* because, even with our highest amount of assumed accretion, they cannot produce stars with masses above $0.73 M_{\odot}$.

Similarly, the amount of transferred mass cannot be *less than 0*. A natural lower bound for the accretion efficiency is then obviously $\varepsilon = 0$. Let's look at the masses of the barium stars that we produce without any RLOF accretion.

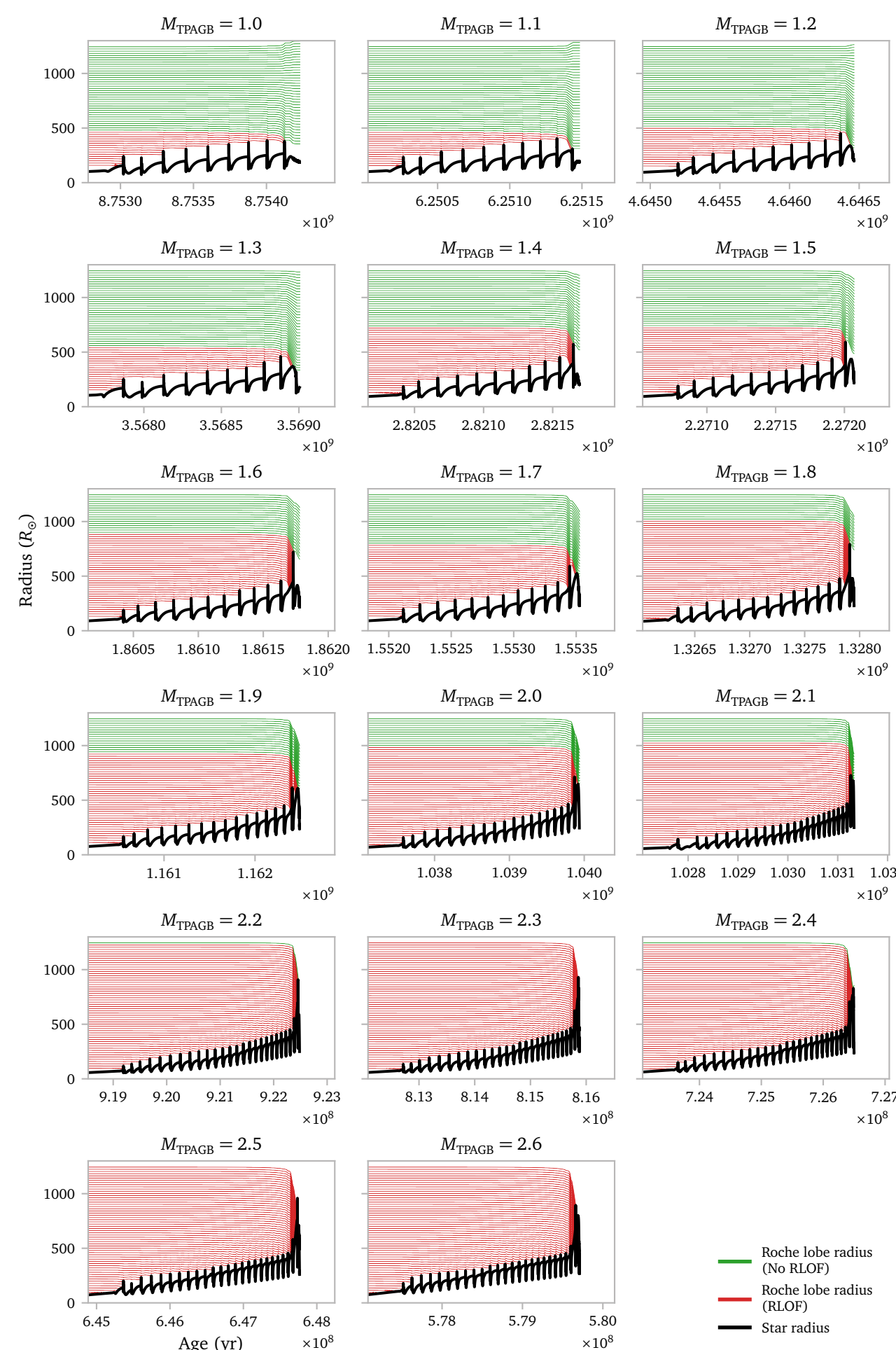


Now we can exclude stars that, even without any RLOF accretion, are heavier than our observed upper limit of $1.5 M_{\odot}$. These are the red regions in the figure above.

Together they form a limit on possible binary configurations. The plots show that this limit is mostly determined by the initial mass ratio, although the initial Roche lobe radius does alter the limits a little bit once winds start to be important.

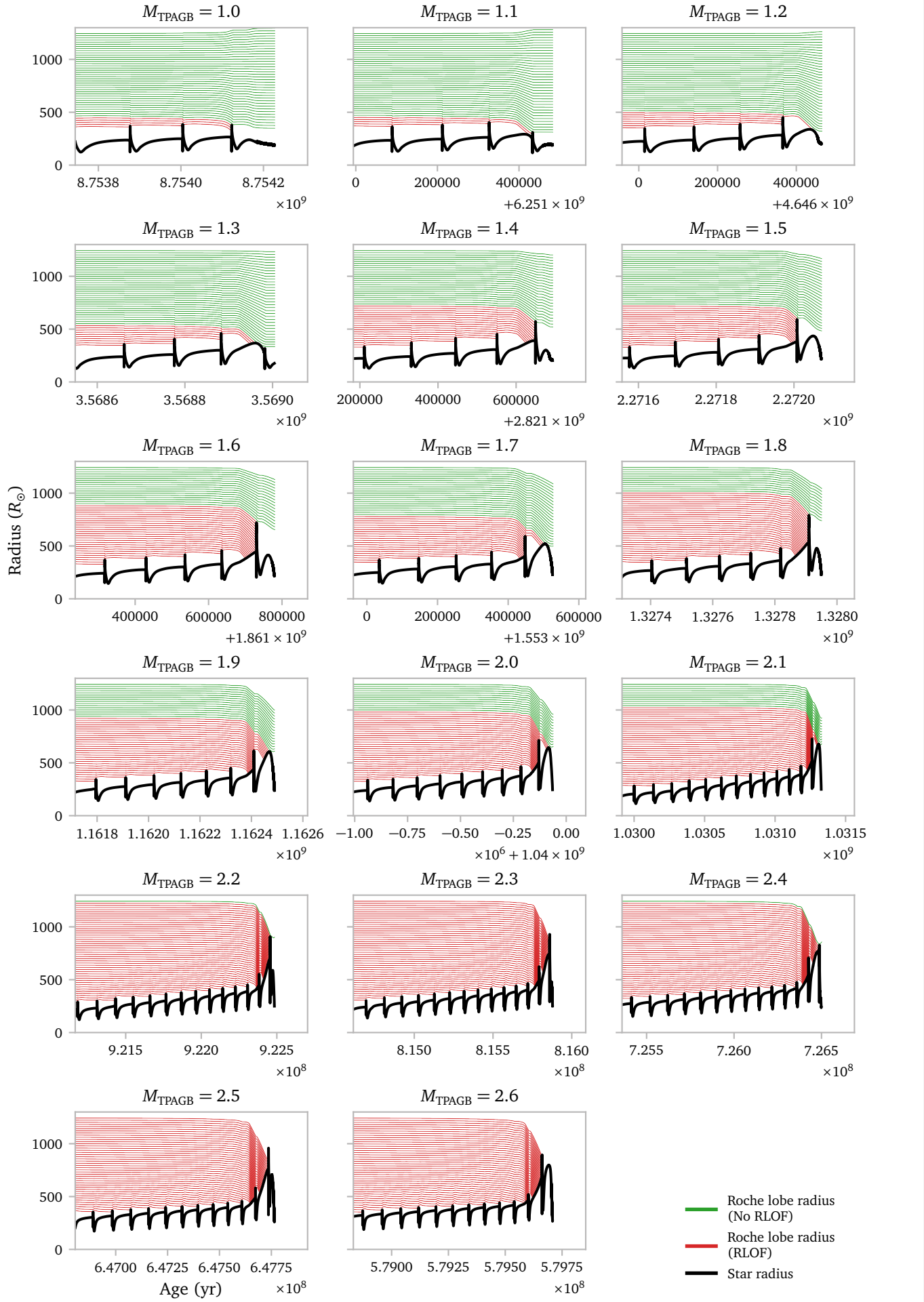
The figures show another thing: namely the boundary between the systems that interact and undergo RLOF and systems that miss RLOF altogether.

To illustrate this more clearly, I show the evolutionary paths for a constant q of 0.8 .



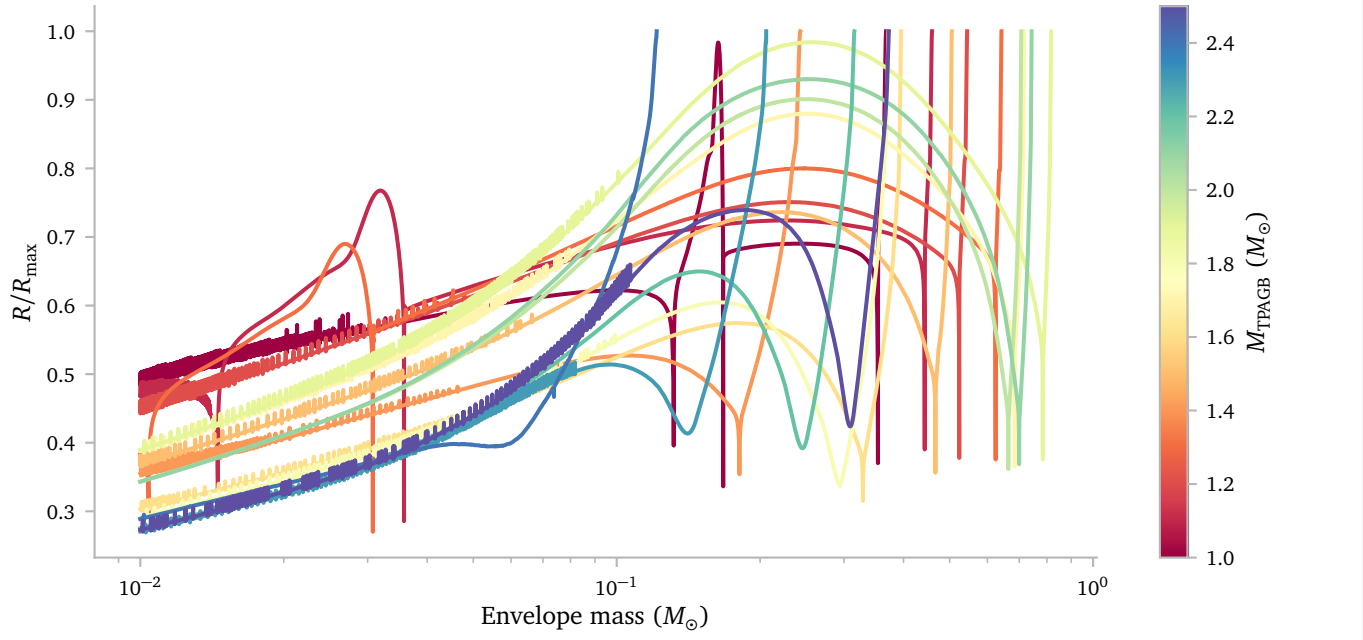
The green lines show the evolution of the Roche Lobe radius for systems that *do not* undergo RLOF. The red lines show systems that *do* undergo RLOF.

Since the most interesting parts seem to occur towards the end, I show the same figure again, but now zoomed in on last few pulses.



For some masses, even a relatively equal mass ratio of 0.8 can cause RLOF interactions *after* the last pulse. This depends on the mass of the convective envelope during the last pulse.

For the 1.7 and 1.9 $M_{\odot}$ TPAGB star, we see a relatively large final bump, while the 1.6 and 1.8 $M_{\odot}$ stars for example, have a much less pronounced final bump.

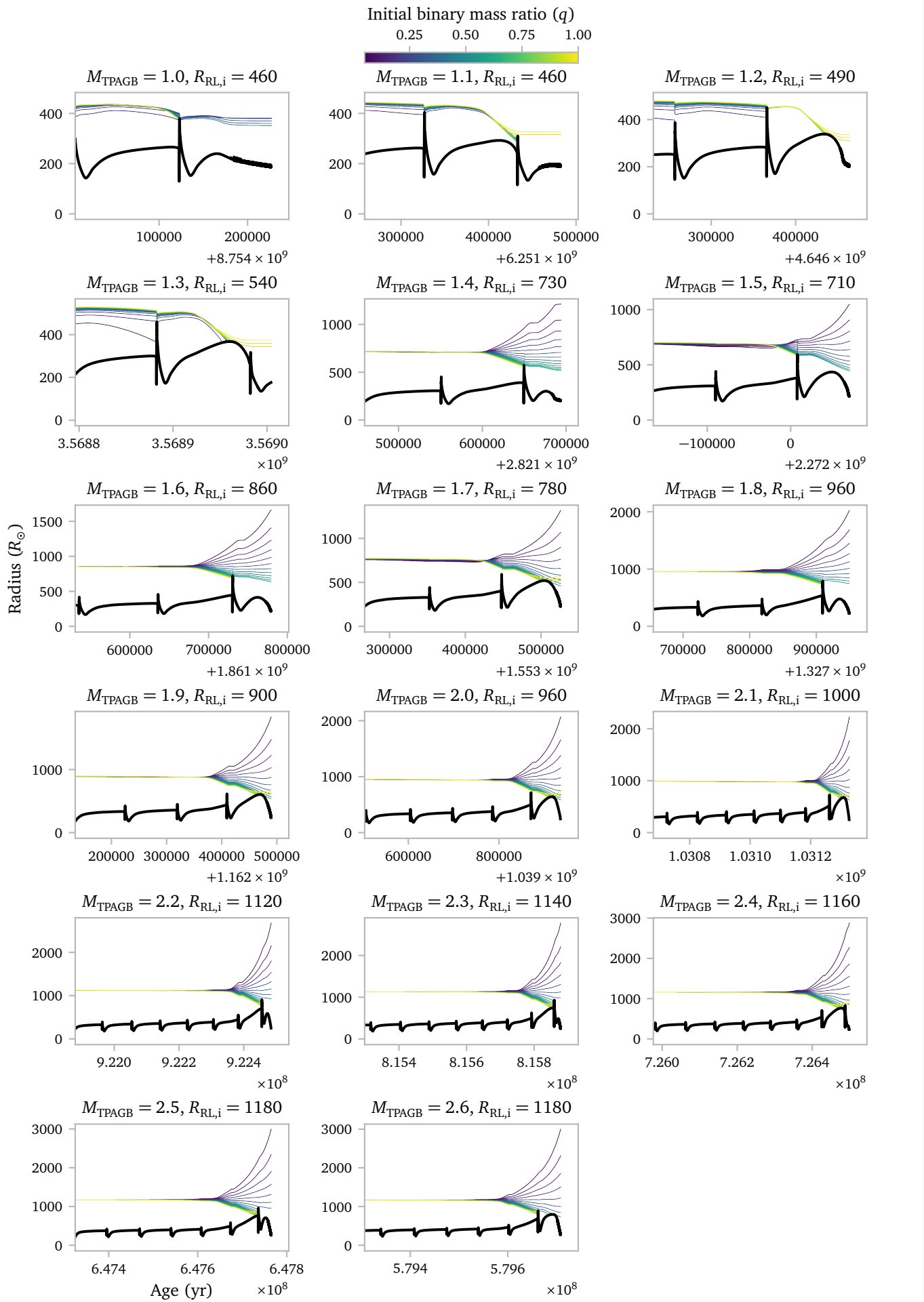


The figure above shows exactly this. It is *not* an effect of initial mass. It is really based on the mass left during the last pulse.

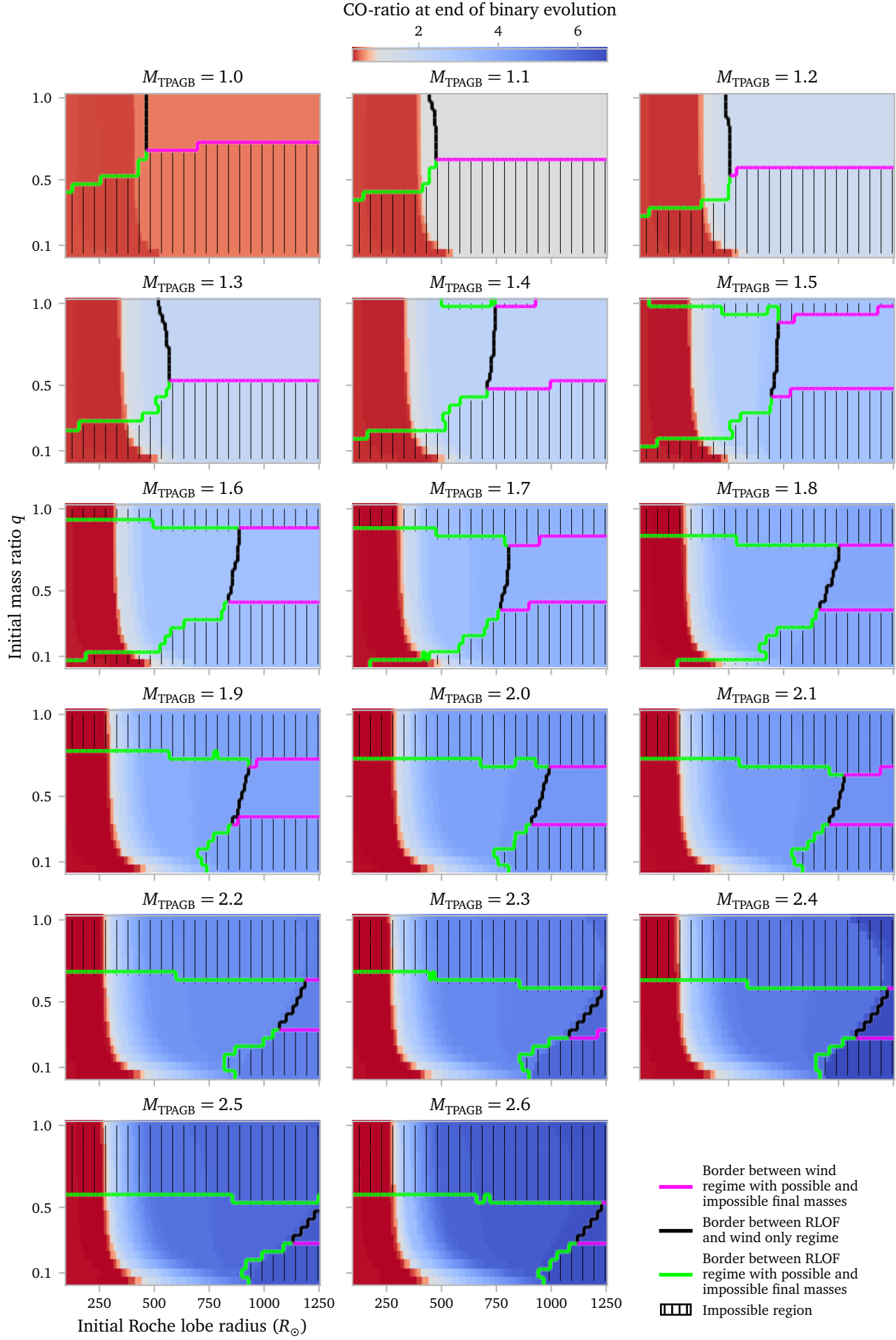
NOTE: I have only *completed* the *MESA* simulations for masses up to 2.2 $M_{\odot}$. A simple calculation however shows that even for masses up to as high as 3.75 $M_{\odot}$, there are still possible configurations with little accretion that *could* form an accreting star with a final mass smaller than 1.5 $M_{\odot}$. I therefore think it might be interesting to extend the single star models to higher masses.

NOTE: I have now simulated all masses up to 2.6 and I am working towards 3.0 $M_{\odot}$.

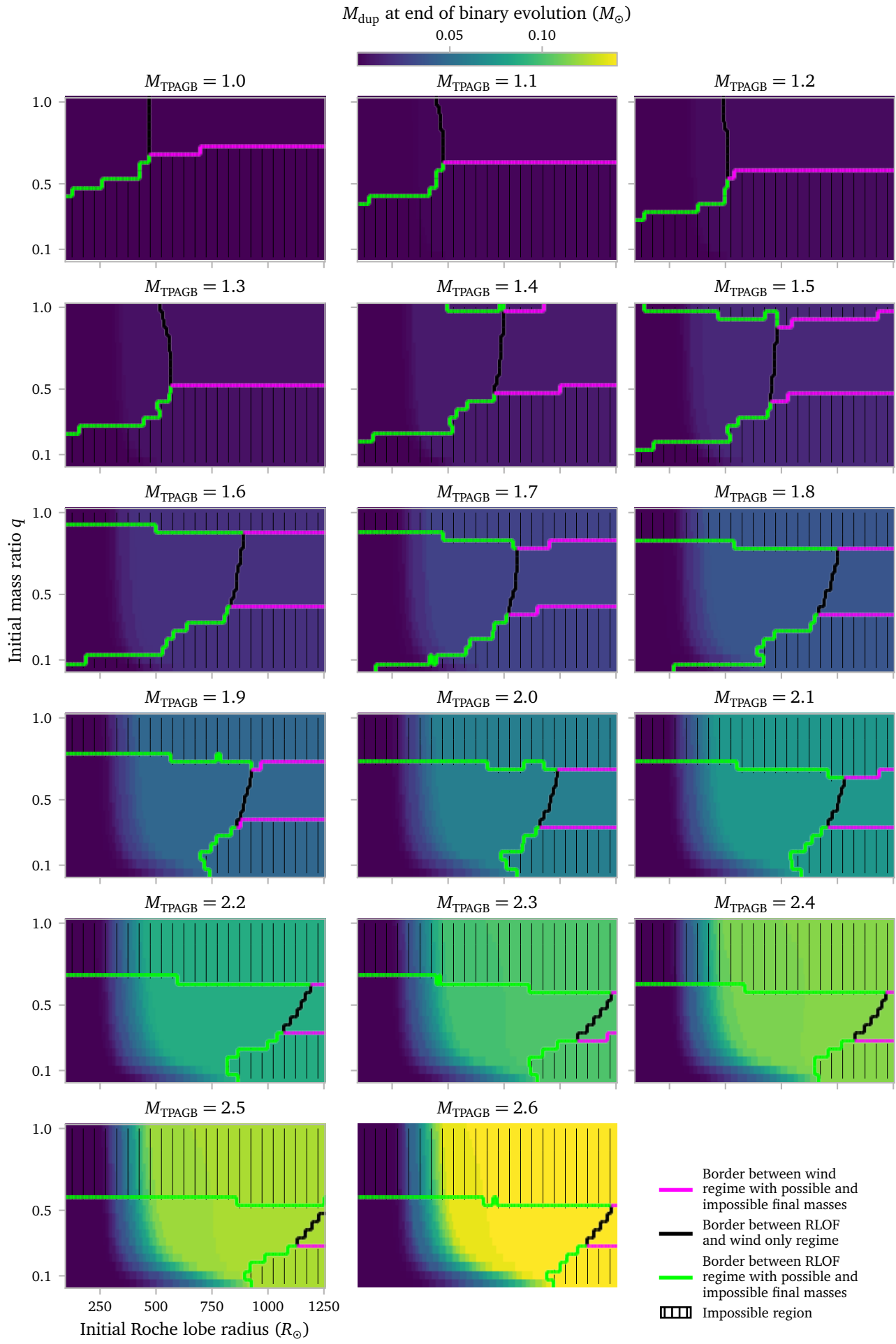
We can also inspect the effects of q . In particular, I think it is interesting to see that the border between the binaries that do and do not interact seems to change direction. The figure below explores this idea.



Now that I have established regions of possible and impossible binary configurations, we can take a look at the other relevant parameters in this grid. Firstly, let me plot the C/O-ratio.



There is a pretty clear divide throughout the mass range, where initial Roche lobe radii of $\lesssim 400 R_{\odot}$ all go into RLOF with a C/O-ratio that is below 1.

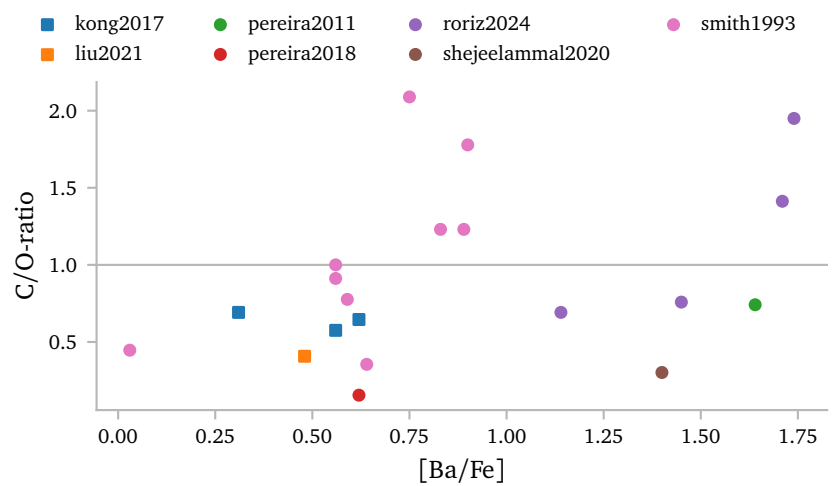


d. Barium dwarf abundances

The barium dwarfs in the Ba_star_orbits.csv unfortunately do not contain C/O-ratios or carbon and oxygen abundances.

Some barium stars do have [Ba/Fe] abundances. They also have a metallicity. In the appendix I compare these values.

I compiled some observation of abundances of barium *dwarfs*³. Below, I plot the C/O-ratio as a function of the barium abundance.

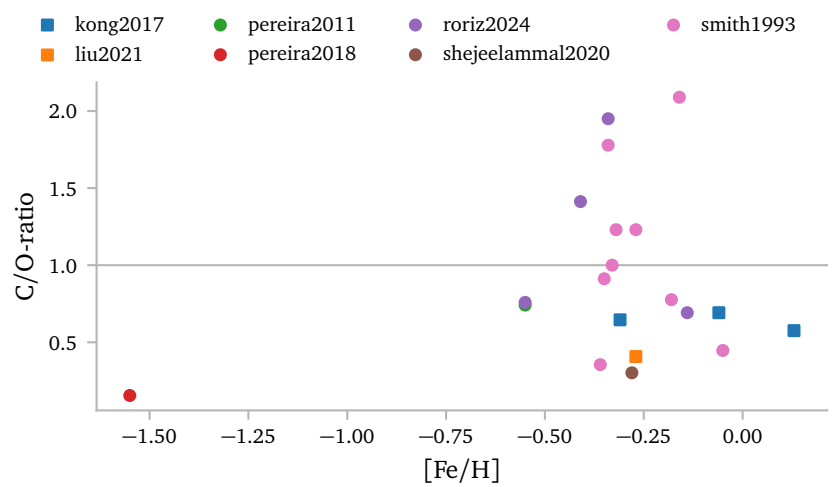


3 and in some cases CH stars.

Roriz et al. (2024); Smith et al. (1993); Pereira & Junqueira (2003); Pereira (2005); Pereira et al. (2019); Kong et al. (2018); Shejeelammal et al. (2020); Liu et al. (2021)

It *seems* like the C/O-ratio *alone* is not a good metric for the barium abundance. There are even some barium dwarfs that have C/O-ratios that are *lower* than the C/O-ratio of my metal-poor MESA main-sequence models.

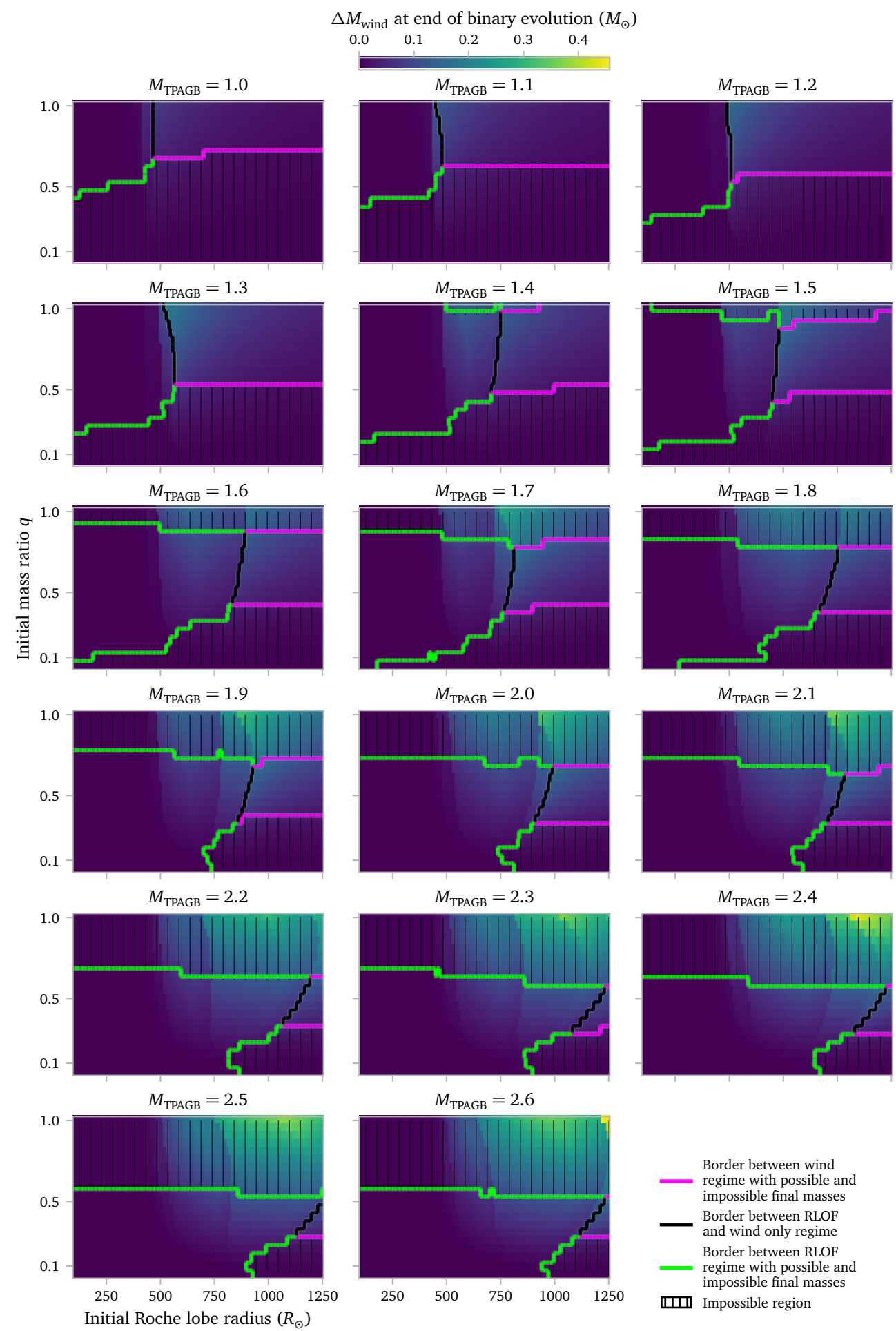
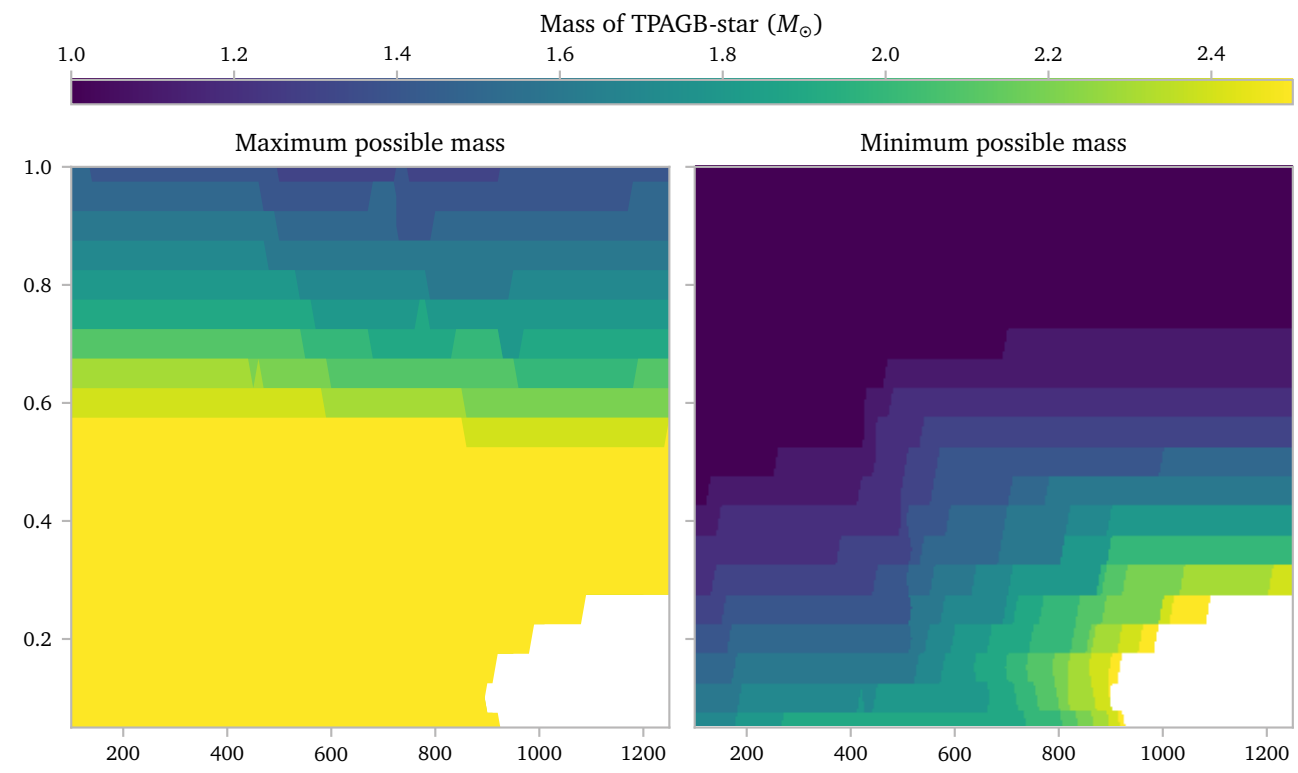
Below I show the C/O-ratio as a function of metallicity.



The red dot seems like an outlier here. Its metallicity is much lower than the other stars, *but*, compared to many more barium dwarfs, the ones that I have not collected abundances for, it is not extreme(Pereira et al., 2019).

I think it is difficult to say anything meaningful based on the C/O-ratio of the accretor.

In part, I think this can be because quite heavy TPAGB stars can have low C/O-ratios due to the hot bottom burning, while still having dredged up mass and probably barium and other s-process elements.



e. Bug fix

In some very rare cases, *MESA* saves a timestep twice with the same exact stellar configuration. This is not a problem, *except* when using the python `evolve.py` code to predict the future of a binary.

This breaks in various places when $\Delta t = 0$.

There is a pretty easy fix for this:

```
def evolve_orbit(
    ...
    while (Bin.status == "detached") & (i < len(dM)):

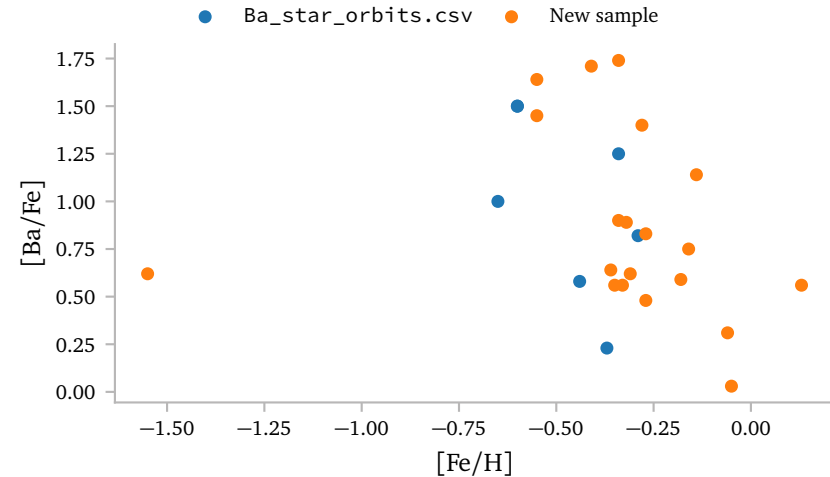
        # Zero-duration stellar timestep: nothing
        ↪ evolves.
    if dt[i] == 0:
        Bin.age.append(Star.age[i + 1])
        Bin.a.append(sma)
        Bin.e.append(ecc)
        Bin.spin1.append(spin)
        Bin.type1.append(Star.type[i + 1])
        Bin.m1.append(Star.mass[i + 1])
        Bin.m2.append(mcomp)
        Bin.amloss.append(amloss)
        Bin.beta.append(0)
        Bin.eta.append(0)
        Bin.vw_over_vorb.append(0)
        Bin.fconv.append(0)

        track_int_simple.append(i)
        i += 1
        continue
    ...
    return Bin
```

References

- Kong, X. M., Bharat Kumar, Y., Zhao, G., et al. (2018) *Three new barium dwarfs with white dwarf companions: BD+68°1027, RE J0702+129 and BD+80°670*. [MNRAS474, 2129](#).
- Liu, S., Wang, L., Shi, J.-R., et al. (2021) *Chemical abundances of three new Ba stars from the Keck/HIRES spectra*. [Research in Astronomy and Astrophysics21, 278](#).
- Pereira, C. B. (2005) *Spectroscopic Verification of Barium Dwarf Candidates: The Analysis of HD 8270, HD 13551, and HD 22589**. [The Astronomical Journal129, 2469](#).
- Pereira, C. B., Drake, N. A., & Roig, F. (2019) *The s-process enriched star HD 55496: origin from a globular cluster or from the tidal disruption of a dwarf galaxy?*. [MNRAS488, 482](#).
- Pereira, C. B. & Junqueira, S. (2003) *Spectroscopic analysis of two CH sub-giant stars: HD 50264 and HD 87080*. [A&A402, 1061](#).
- Roriz, M. P., Holanda, N., da Conceição, L. V., et al. (2024) *High-resolution Spectroscopic Analysis of Four Unevolved Barium Stars*. [AJ167, 184](#).
- Shejeelammal, J., Goswami, A., Goswami, P. P., Rathour, R. S., & Masseron, T. (2020) *Characterizing the companion AGBs using surface chemical composition of barium stars*. [MNRAS492, 3708](#).
- Smith, V. V., Coleman, H., & Lambert, D. L. (1993) *Abundances in CH Subgiants: Evidence of Mass Transfer onto Main-Sequence Companions*. [ApJ417, 287](#).

A Additional plots



It seems, at least with this relatively small sample, both groups draw from the same distribution.